ALKENE

61. The compound having both *sp* and sp^2 hybridised carbon atom is

(a) Propene

(b) Propyne

(c) Propadiene

(d) None of these

62. The halogen which is most reactive in the halogenation of alkenes under sunlight is

(a) Chlorine (b) Bromine

(c) Iodine (d) All equal

63. When ethene reacts with bromine, it forms

(a) Chloroethane

(b) Ethylene dibromide

(c) 1 bromopropane

(d) 1,2-dichloroethene

64. Paraffins are soluble in

(a) Distilled water (b) Benzene

(c) Methanol (d) Sea water

65. Addition of *HCl* to propene in presence of peroxides gives

(a) 1-Chloropropane

(b) 2-Chloropropane

(c) 3-Chloropropane

(d) Propene dichloride

66. The name of the product obtained by the addition of HI to propene in presence of peroxide catalyst is

(a) Isopropyl iodide

(b) 2-Iodopropene

(c) 2-Iodopropane

(d) 1-Iodopropane

67. In the reaction $C_2H_5CH = CH_2 + H - X \rightarrow$ Product. What is the product

(a) $C_2H_5 - CH_3$

(b) $C_2H_5CH_2 - CH_2X$

(c) $C_2H_5 - CHX - CH_3$

(d) $CH_3 - CH_2X - CH = CH_2$

68. Alkene can be prepared from alkyl halide by the following reagent $R - X + Nu^- \rightarrow \text{Alkene} + NuH$

(a) Alc. *KOH* + heat

(b) *Aq*. *KOH* + cold water

(c) *NaOH*

(d) *LiOH*

69. 2-chlorobutane is heated with alcoholic *NaOH*, the product formed in larger amount is

(a) 1-Butene (b) 1-Butyne

(c) 2-Butene (d) All of these

70. Ethylene has high b.p. and high vapour pressure at $100^{o}C$ and does not dissolve in water. Hence ethylene is separated by this method

(a) Simple distillation

(b) Vacuum distillation

(c) Vapour distillation

(d) Alkali treatment

71. Addition of bromine to 1, 3-butadiene gives

(a) 1, 4-addition product only

(b) 1, 2-addition product only

(c) Both 1, 2-and 1, 4 addition product

(d) None of these

72. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikoff's addition to alkenes because

(a) Both are highly ionic

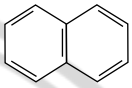
(b) One is oxidising and the other is reducing

(c) One of the steps is endothermic in both the cases

(d) All the steps are exothermic in both the cases

73. The compound most likely to decolourize a solution of potassium permanganate is

(a) CH_3CH_3

(b) 

(c) $CH_3CH = CHCH_2CH_3$

(d) $CH_3 - \overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}} - CH_3$

74. Ethylene is converted to *X* on passing through a mixture of an acidified aqueous solution of palladium chloride and cupric chloride. Which of the following reagents readily take part in addition reaction with *X*

(a) Br_2 (b) *HBr*

(c) *HCl* (d) *HCN*

75. Addition of *HCl* does not obey antimarkownikoff's rule because

(a) It is a strong acid

(b) It is a gas

(c) Its bond energy is high

(d) Its bond energy is less

76. Correct statement about1, 3-dibutene

(a) Conjugated double bonds are present

(b) Reacts with *HBr*

(c) Forms polymer

(d) All of these

77. At low temperatures, the slow addition of molecular bromine to $CH_2 = CH - CH_2 - C \equiv CH$ gives

(a) $CH_2 = CH - CH_2 - CBr = CHBr$

(b) $BrCH_2 - CHBr - CH_2 - C \equiv CH$

(c) $CH_2 = CH - CH_2 - CH_2 - CBr_3$

(d) $CH_3 - CBr_2 - CH_2 - C \equiv CH$

78. PCl_5reacts with propanone, to give

(a) *vic*-dichloride

(b) Propanal

(c) propane-chloride

(d) gem-dichloride

79. The compounds that will give an isomer of 2; 2-dimethyl propane on catalytic hydrogenation are

(1) $CH_3CH = \underset{\underset{CH_3}{|}}{C} - CH_3$

(2) $CH_3CH = CHCH_3$

(3) $CH_3\overset{\overset{H}{|}}{C} = CHCH_2CH_3$

(4) $CH_3\underset{\underset{CH_3}{|}}{C} = \underset{\underset{CH_3}{|}}{C} - CH_3$

(a) 1 and 4 (b) 2 and 4

(c) 1 and 3 (d) 1 and 2

80. Alkene $R - C - H = CH_2$ reacts readily with B_2H_{56} and the product on oxidation with alkaline hydrogen peroxides produces

(a) $R - CH_2 - CHO$

(b) $R - CH_2 - CH_2 - OH$

(c) $R - \underset{\underset{O}{||}}{C} - CH_3$

(d) $R - \underset{\underset{OH}{|}}{CH} - \underset{\underset{OH}{|}}{CH_3}$